

K1388 — the HARD auditor call + two theorem-grounded advances. ★ Casey’s DESI-reversal (“the negative curvature is a constant; DESI reads the relaxing continuum BACKWARDS — inward-from-infinity instead of BST’s outward projection; the w_a sign is an orientation artifact”) is a LEGITIMATE, corpus-consistent CANDIDATE RESOLUTION — but it is ALSO structurally identical to the falsifier-escape we all forbade yesterday, so it resolves NOTHING until the outward orientation FORCES the w_a sign BLIND. Until then the falsifier STANDS: BST $w_a > 0$ vs DESI $w_a < 0$ at $\sim 3\sigma$, BST on the wrong side. I challenge the PI here (equal standing) with the insight taken seriously. ★ Ratify Grace’s TWO-THEOREM grounding of the polarity/phase ($32 \rightarrow 16$ = drop the $SO(2)$ phase circle; boundary is maximal-totally-real = keeps real, drops phase) — elevates K1387 from heuristic to theorem-grounded. ★ Ratify Lyra’s algebra-vs-module reframe: 16 (Clifford algebra = capacity = cell-count) and 4 (spinor module = states = the ± 4 edge index) are TWO FACES of ONE Clifford — the 16-vs-4 is not a competition. Bank Cal’s count-once. Cal cold-reads the DESI bar.

Keeper (2026-08-11, $\sim 12:40$. Casey answered the DE falsifier with a reversal that would flip the sign. It’s corpus-consistent AND it’s the exact shape of the laundering we banned yesterday. The only honest path is a pre-committed blind forcing test. I hold the line — respectfully, and taking the physics seriously. Cal cold-reads. Nothing pushed.)

★ THE HARD CALL — Casey’s DESI-reversal: candidate resolution, NOT a falsifier escape (yet)

Casey: *the negative curvature of D_{IV}^5 with the boundary is a constant; DESI’s w_a is an approximation of the relaxing continuum; they view it backwards — projecting inward from infinity when we project outward.*

Why it’s legitimate (I take it seriously): - It’s corpus-consistent, not invented for the occasion. BST’s banked ontology is pure positive-time, arrow dynamical-not-geometric, outward/bubble-nucleation (project_bst_pure_positive_time_ontology; K1382). “Outward projection” is not ad hoc — it’s the established BST arrow. So a sign subtlety between BST’s intrinsic (outward, boundary-relaxing) $w(a)$ and the observer’s CPL fit (which assumes a particular orientation) is a *physically real question*, not a dodge. - Constant negative curvature = a geometric $w = -1$ with the apparent w_a being the projection of the relaxation — coherent.

Why it resolves NOTHING yet (I hold the line — equal standing, per your mandate): - This is structurally identical to the falsifier-escape we ALL forbade yesterday (K1387; Elie + Lyra + me: “Casey’s cosmology doesn’t get to invent a friendlier $w(a)$; the day we launder that is the day the rest stops being worth anything”). A reinterpretation that flips the sign to match the data is laundering *regardless of how elegant it is* — and it is most dangerous when it’s the PI proposing it and the whole room wants it to be true. - The discriminator is FORCING, blind and pre-committed: Does the BST-forced outward orientation *yield* $w_a < 0$ when the breathing-mode w_a is re-derived from that orientation, blind, before comparison to DESI? And did the original $w_a > 0$ computation assume INWARD (a genuine sign error), or outward (in which case Casey’s point doesn’t bite and the falsifier stands)? - FORCED + blind + it flips $\rightarrow$ legitimate correction of an orientation error; the falsifier becomes a *confirmation*, and it’s a major result. - CHOSEN to match $\rightarrow$ laundering, forbidden. - Until that re-derivation forces the sign, the falsifier STANDS: BST $w_a > 0$, DESI DR2 $w_a < 0$ at $\sim 3\sigma$, parametrization-robust — BST is on the wrong side of the data. The existence of a candidate resolution does NOT soften that sentence. We do not get to feel resolved

because a resolution is imaginable.

The concrete blind test (route it): (1) Re-derive the breathing-mode $w(a)$ sign with the projection orientation BST's positive-time ontology *forces*, blind to DESI, pre-committed. (2) Independently audit the ORIGINAL $w_a > 0$ derivation: which orientation did it assume? If it assumed inward and BST forces outward, that's a real bug and the correction is honest. (3) Cal + I hold the pre-committed bar: whatever the blind re-derivation gives, we report — no adjusting to match. This is a genuine, high-value forward computation — it could turn our worst falsifier into our sharpest confirmation, but ONLY if earned blind.

★ Ratify Grace's TWO-THEOREM grounding of the polarity/phase (K1387 elevated: heuristic → theorem)

I verified both anchors: 1. The boundary stores exactly half — by the maximal-totally-real theorem. D_{IV}^5 is complex-dim 5 = real-dim 10; its Shilov boundary $(S^4 \times S^1)/\mathbb{Z}_2$ is real-dim 5 = exactly half. This is the Shilov boundary being maximal totally real — it keeps the real slice and drops the imaginary. “A reduction stores the polarity and lets the phase go” IS the statement that the boundary is totally real. Casey described a theorem in a computer scientist's words. My K1387 flagged “reduction-stores-polarity needs rigor” — Grace supplied the rigor. 2. The dropped half is specifically the phase. The boundary S^1 is $SO(2)$ — the phase/time circle (the $C/U(1)$ summand, banked). $2^5=32$ factors as S^4 polarity ($2^4=16$) $\times$ S^1 phase ($\times 2$); $32 \rightarrow 16$ is dropping the $SO(2)$ phase circle, verified ($16 \times 2 = 32$). Not “halve because boundaries halve” — “drop the $SO(2)$ phase, keep the spatial orientation.” Mechanism, not coincidence. - [?] The $32 \rightarrow 16$ step is now theorem-grounded. What remains open is only the 16-vs-4 (below).

★ Ratify Lyra's algebra-vs-module reframe (the 16-vs-4 was never a competition)

Lyra's insight, verified: 16 and 4 are the two natural numbers of ONE Clifford algebra $Cl(\mathbb{R}^4)$: - $16 = 2^4$ = the algebra dimension — the CAPACITY, one polarity bit per direction = the cell-count = DM/gravity. - $4 = 2^{d/2}$ = the spinor module dimension — the STATES the algebra acts on = the ± 4 edge index (the $SU(2)_L$ doublets). - Same Clifford, two faces. So handedness (edge index = module), the dark shadow, and gravity's weakness (cell-count = algebra) are ONE algebra's faces, graded by ONE switch (K825 chirality $\mathbb{Z}_2$). This is Cal's count-once made concrete AND it dissolves the 16-vs-4 tension: the cell-count reads the *algebra*, the edge index reads the *module* — both real, both from one Clifford. - The one nail left (Lyra's): WHY does the cell-count (what's stored/observed) read the *algebra* (16) while the index reads the *module* (4)? Casey's measurement principle answers it — *to observe is to commit to the polarity (the algebra grading), leaving the conjugate phase (and the module amplitude) in the bulk*. Reason, not fit — but “stored quantity = the algebra grading” must be made rigorous, blind. The 16-vs-4 is now a principled split, not an ambiguity; the forcing is why-stored=algebra.

Bank Cal's count-once (unchanged, reinforced)

The edge-sign χ_{int} (odd-dim V_{12} : the central volume element $\gamma_1\gamma_2\gamma_3 = \pm i$ distinguishing the two $Cliff(3)$ modules) IS the K825 switch. Edge-sign + cell-count = ONE switch; a dual PASS = ONE confirmation, not two. And “three routes $\rightarrow 16$ ” (Casey polarity / Grace Clifford / Lyra sign-bit) is ONE structure in three vocabularies — corroboration, counts once (overdetermined $\neq$ votes). Edge sign = do color-triplet (quark) and color-singlet (lepton) sit in the same $Cliff(3)$ module ($\rightarrow SM$) or opposite ($\rightarrow mirror$); $SU(3)$ vector-like preserves χ_{int} within an irrep but doesn't fix triplet-vs-singlet. Genuinely open. $d=10 \equiv 2$ confirmed.

Route

1. ★ Elie + Lyra — the DESI blind forcing test (NEW, high-value): re-derive the breathing-mode $w(a)$ sign from BST's forced outward/positive-time orientation, blind + pre-committed; audit whether the original $w_a > 0$ assumed inward (a sign bug). Verify current DESI DR2 numbers first. Cal + Keeper hold the pre-committed bar. Until it forces, the falsifier STANDS.
2. ★ Lyra + Grace — the one remaining count nail: why does the cell-count read the Clifford *algebra* (16) and the edge index the *module* (4)? Casey's measurement principle (store the polarity) is the candidate — make

“stored = algebra grading” rigorous, blind ($32 \rightarrow 16$ is now theorem-grounded via the $SO(2)$ -phase drop; only this is left). Then $\div N_c$ is honest (3 colors).

3. Cal + Elie — the edge sign: the $Cliff(3)$ module assignment (triplet vs singlet), blind; + Toeplitz KO-degree at KO_0 ($d=10$). Same K825 switch — count once.
4. DE falsifier — banked live; Casey’s reversal = candidate resolution held to (1). Do not launder. 137 stays in the 7-bit layer.
5. Keeper — recused from the counts (K1377); verified the two theorems, framed the DESI blind test, hold the pre-committed bars (K1377 + the DESI orientation bar).

— Keeper, K1388, 2026-08-11. HARD CALL: Casey’s DESI-reversal (negative curvature constant; DESI reads the relaxing continuum inward-from-infinity, BST projects outward; w_a sign = orientation artifact) is LEGITIMATE + corpus-consistent (BST positive-time/outward ontology, `project_bst_pure_positive_time_ontology`, K1382) — but structurally IDENTICAL to the falsifier-escape banned yesterday (K1387). Resolves NOTHING until the outward orientation FORCES the w_a sign BLIND + pre-committed (re-derive breathing-mode w_a from forced orientation; audit if original $w_a > 0$ assumed inward = a bug). FORCED+blind+flips $\rightarrow$ legit correction (falsifier $\rightarrow$ confirmation, major); CHOSEN-to-match $\rightarrow$ laundering, forbidden. Until forced, falsifier STANDS: BST $w_a > 0$ vs DESI $w_a < 0 \sim 3\sigma$, BST wrong side; a candidate resolution does NOT soften it. Equal-standing challenge to the PI. RATIFY Grace 2 theorems ($32 \rightarrow 16$ theorem-grounded): (1) boundary real-dim 5 = half of 10 = maximal-totally-real (keeps real/polarity, drops imaginary/phase) — “reduction stores polarity” IS “boundary is totally real”; (2) dropped half = $SO(2)$ phase circle ($S^4 \times S^1$)/ $\mathbb{Z}_2$, $32=16 \times 2$, drop the phase keep spatial orientation. RATIFY Lyra algebra-vs-module: $16=Cl(R^4)$ algebra dim (capacity=cell-count=DM/gravity), $4=2^{d/2}$ spinor module (states= ± 4 edge index) — TWO FACES of ONE Clifford, 16-vs-4 not a competition; nail = why cell-count reads algebra (Casey measurement principle: store polarity, rigor owed). Bank Cal count-once (edge+cell = one K825 switch, dual PASS=one confirmation; 3 routes=one structure). Route: Elie/Lyra DESI blind forcing (falsifier stands till forced); Lyra/Grace why-stored=algebra; Cal/Elie edge $Cliff(3)$ module; DE banked live; Keeper recused, holds bars. Cal cold-reads. Nothing pushed.